

Formal Solutions of Linear Differential Systems with Essential Singularities in their Coefficients

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ABSTRACT

The local analysis of formal meromorphic linear differential systems with coefficients in $\mathbb{C}((z))$ has been widely studied in the literature and there exist various computer algebra algorithms for computing formal solutions of such systems. In the present paper we extend the algorithm presented in [3] to allow more general systems. More precisely, we give an algorithm for computing a formal fundamental matrix of solutions around $z = 0$ of systems with coefficients in $\mathbb{C}((z))[[X]]$, where X is transcendental and hyperexponential over $\mathbb{C}((z))$.

Categories and Subject Descriptors

I.1.2 [Symbolic and Algebraic Manipulation]: Algorithms

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Computer Algebra, Algorithms, Linear Differential Systems with Essential Singularities, Hyperexponential Extensions, Formal Reduction

1. INTRODUCTION

Let z be a complex variable and $\mathbb{C}[[z]]$ (resp. $\mathbb{C}((z))$) the ring (resp. field) of formal power (resp. Laurent) series in z with coefficients in the field $\mathbb{C}$ of complex numbers. Let $\mathbb{M}_n(\mathbb{C}[[z]])$ be the ring of square matrices of size $n \in \mathbb{N}^*$ with entries in $\mathbb{C}[[z]]$. In the present paper, we consider formal linear differential systems of the form

$$\frac{dY}{dz} = z^{-p} A(z, X) Y, \quad p \in \mathbb{N}^*, \quad (1a)$$

$$\frac{dX}{dz} = z^{-q} a(z) X, \quad q \geq 2, \quad a(z) \in \mathbb{C}[[z]], \quad a(0) \neq 0, \quad (1b)$$

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where $A(z, X) = \sum_{k=0}^{+\infty} A_k(z) X^k$ with $A_k(z) \in \mathbb{M}_n(\mathbb{C}[[z]])$ and $Y = (y_1, \dots, y_n)^T$ is a vector of unknown functions.

The conditions on the integer q and the formal power series $a(z)$ imply that $X = \exp(\int z^{-q} a(z) dz)$ is transcendental over $\mathbb{C}((z))$ (see, for example, [20, Theorem 2.4]). The coefficients of System (1a) then have an essential singularity at the origin since $z = 0$ is neither a removable singularity nor a pole.

Linear differential systems with coefficients in hyperexponential extensions appear in many applications as for example the linearization of a dynamical system defined over a base field F around a particular exponential solution ϕ . Indeed the resulting linear variational equations are linear differential systems with coefficients in $F(\phi)$ (see [1] and references therein). Moreover the study of linear differential systems with coefficients in hyperexponential extensions can be used to compute a closed form of an integral (see [6]). In computer algebra, linear differential systems (equations) with coefficients in hyperexponential extensions have been studied in [7, 8, 9]. In [10, 11], the author develops algorithms for computing closed form solutions (e.g., polynomial, rational solutions) of scalar linear differential equations with coefficients in hyperexponential extensions. More recently, the latter algorithms have been extended to handle systems directly and applied to indefinite integration in [6].

The systems (1) considered in the present paper can be viewed as a generalization of formal meromorphic linear differential systems of the form

$$\frac{dY}{dz} = z^{-p} A(z) Y, \quad p \in \mathbb{N}^*, \quad (2)$$

where $A(z) \in \mathbb{M}_n(\mathbb{C}[[z]])$ and $A(0) \neq 0$. The local analysis of formal meromorphic systems of the form (2) has been well investigated: see, e.g., [19, 18, 21, 2, 14, 3, 5, 4] and references therein. In [3], the first author gave an algorithm for computing a fundamental matrix of formal solutions of (2) of the form

$$Y = \Phi(t) t^\Lambda \exp(Q(1/t)), \quad (3)$$

with $z = t^r$ where $r \geq 1$ is a positive integer called the *ramification index*, $Q(1/t) = \text{diag}(q_1(1/t), \dots, q_n(1/t))$ is a diagonal matrix containing polynomials in $1/t$ without constant terms, Λ is a constant matrix commuting with Q , and $\Phi(t)$ is a matrix of formal Laurent series in t .

The main contribution of the present paper is to provide an algorithm for computing a formal fundamental matrix of

solutions of (1) of the form

$$Y = \left(\sum_{k=0}^{+\infty} \Phi_k(t) X^k \right) t^\Lambda \exp(Q(1/t)), \quad (4)$$

with the notation of (3) and $\Phi_k(t) \in \mathbb{M}_n(\mathbb{C}((t)))$, for all $k \in \mathbb{N}$. Note that in the particular case $X = \exp(1/z)$, a similar form of a formal fundamental matrix of solutions can be deduced from the work of M. Bouffet in [9].

Our approach consists in viewing (1a) as a *perturbation* of the meromorphic system obtained by letting $X \rightarrow 0$ in (1a). Several cases have to be distinguished by comparing the integers p and q involved in (1). When $p < q$, we prove that (1) can be reduced via a matrix transformation $T \in \mathbb{M}_n(\mathbb{C}[[z]][[X]])$ to the formal meromorphic system $Y' = z^{-p} A_0(z) Y$. In the case $p = q$, one must investigate the spectrum of the leading matrix $A_0(0)$. We prove that up to an invertible transformation in $\mathbb{M}_n(\mathbb{C}[[z]][[X]])$, we can assume that the eigenvalues of $A_0(0)$ do not differ by an integer multiple of $a(0)$ and in this case, similarly to the case $p < q$, we prove that (1) can be reduced to $Y' = z^{-p} A_0(z) Y$. The last case $p > q$ is much more involved: by introducing, if necessary, ramifications of the form $z = t^r$, we prove that we can apply a (finite) sequence of transformations in order to reduce the problem to the case where either $p \leq q$ or $n = 1$ (scalar case). Our methods are constructive and a Maple implementation will be soon available.

The paper is organized as follows. In Section 2, we introduce the different rings of coefficients considered in the sequel and we briefly review the algorithm of [3] for computing a formal fundamental matrix of solutions of formal meromorphic systems (2). In Section 3, we introduce the systems (1) considered in this paper and give an overview of our main results. Then, Section 4 handles one by one the different cases that have to be distinguished by our algorithm for computing a formal fundamental matrix of solutions of (1). In Section 5, we discuss extensions of our algorithm to a more general class of differential systems.

2. PRELIMINARIES

2.1 Rings of coefficients

In this subsection, we investigate the problem of defining the most suitable ring of coefficients for systems of the form (1).

Let z be a complex variable, $\mathbb{C}$ the field of complex numbers and consider the ring $\mathbb{C}[[z]]$ of formal power series in z with coefficients in $\mathbb{C}$ and its quotient field $\mathbb{C}((z))$, namely the field of formal Laurent series in z with coefficients in $\mathbb{C}$. The ring $\mathbb{C}[[z]]$ (resp. the field $\mathbb{C}((z))$) is endowed with a differential ring (resp. field) structure by considering the derivation $' = d/dz$. For $f \in \mathbb{C}((z))$, we denote by $v_z(f)$ the usual z -adic valuation of f , i.e., the order of f in z with $v_z(0) = +\infty$. Now let X be a transcendental element satisfying (1b), i.e., $X' = z^{-q} a(z) X$, with $q \geq 2$, $a(z) \in \mathbb{C}[[z]]$, and $a(0) \neq 0$, and consider the ring $\mathcal{R} = \mathbb{C}((z))[[X]]$ of formal power series in X with coefficients in the field of formal Laurent series in z . An element $f \in \mathcal{R}$ can be written as $f = \sum_{k=0}^{+\infty} f_k(z) X^k$, with $f_k(z) \in \mathbb{C}((z))$, and the derivation $' = d/dz$ extends naturally to $\mathcal{R}$ by the formula $f' = \sum_{k=0}^{+\infty} (f'_k(z) + k z^{-q} a(z) f_k(z)) X^k$.

LEMMA 1. *The ring of constants of the differential ring $(\mathcal{R}, d/dz)$ is $\mathbb{C}$.*

PROOF. An element $f \in \mathcal{R}$ satisfies $f' = 0$ if and only if $f'_k(z) + k z^{-q} a(z) f_k(z) = 0$ for all $k \geq 0$. This yields $f_0(z) \in \mathbb{C}$ and for all $k \geq 1$, we get $f_k(z) = c_k X^{-k}$, with $c_k \in \mathbb{C}$. Now by our hypothesis, $q \geq 2$ so that X is a transcendental element over $\mathbb{C}((z))$ (see, for example, [20, Theorem 2.4]). Consequently, since $f_k(z) \in \mathbb{C}((z))$ we necessarily have $f_k(z) = 0$ for all $k \geq 1$ and we finally get $f = f_0(z) \in \mathbb{C}$ which ends the proof. $\square$

The coefficients of systems of the form (1a) have their z -adic valuation bounded below. However, not every system with coefficients in $\mathcal{R}$ can be written in the form (1). Therefore, let us introduce the following subset $\mathcal{A} \subsetneq \mathcal{R}$:

$$\mathcal{A} = \left\{ f = \sum_{k=0}^{+\infty} f_k(z) X^k \in \mathcal{R} \mid \inf_{k \in \mathbb{N}} v_z(f_k(z)) > -\infty \right\}. \quad (5)$$

By a straightforward verification, one can show that $\mathcal{A}$ is a differential subring of $\mathcal{R}$ endowed with the derivation d/dz .

LEMMA 2. *The application $v : \mathcal{A} \rightarrow \mathbb{Z} \cup \{+\infty\}$ defined by $v(f) = \min_{k \in \mathbb{N}} v_z(f_k(z))$ for $f = \sum_{k=0}^{+\infty} f_k(z) X^k \in \mathcal{A}$ is a valuation of $\mathcal{A}$.*

PROOF. It is straightforward to show that, for $f_1, f_2 \in \mathcal{A}$, $v(f_1) = +\infty \Leftrightarrow f_1 = 0$, $v(f_1 + f_2) \geq \min(v(f_1), v(f_2))$ and $v(f_1 f_2) \geq v(f_1) + v(f_2)$. It remains to prove that we have $v(f_1 f_2) = v(f_1) + v(f_2)$. Since $f_1, f_2 \in \mathcal{A}$, we can write $f_i = p_i + z^{v(f_i)} \tilde{f}_i$ with $v(p_i) > v(f_i)$ and $v(\tilde{f}_i) = 0$, $i = 1, 2$. Then we have

$$f_1 f_2 = p_1 p_2 + p_1 z^{v(f_2)} \tilde{f}_2 + p_2 z^{v(f_1)} \tilde{f}_1 + z^{v(f_1)+v(f_2)} \tilde{f}_1 \tilde{f}_2,$$

and inspecting the valuation of each term of the previous sum, we get the result. $\square$

In the process of the algorithm described in Section 4, even if we start with a system with coefficients in the ring $\mathcal{A}$, it may happen, at some stage, that we get a new system whose coefficients do not belong to $\mathcal{A}$. Typically, one can obtain the formal power series $\sum_{k=0}^{+\infty} z^{-k} X^k$. Consequently, we need to relax a bit the condition $\inf_{k \in \mathbb{N}} v_z(f_k(z)) > -\infty$ and introduce a bigger ring.

Let $f = \sum_{k=0}^{+\infty} f_k(z) X^k \in \mathcal{R}$. For $k \in \mathbb{N}$, we consider the vertical half-line $l_k = \{(k, y) \mid y \geq v_z(f_k)\} \subset \mathbb{R}^2$ with $l_k = \emptyset$ if $f_k = 0$. Then we define the *Newton polygon* $\mathcal{N}(f)$ of f to be the boundary of the convex hull of $\cup_{k \in \mathbb{N}} l_k$ (see, for example, [15]). The Newton polygon is said to be *degenerate* if it is reduced to the y -axis. We now define a new ring $\mathcal{B} \supsetneq \mathcal{A}$ as follows:

$$\mathcal{B} = \{f \in \mathcal{R} \mid \exists \alpha, \beta \in \mathbb{Q}; \forall k, v_z(f_k) \geq \alpha k + \beta\}. \quad (6)$$

The elements of $\mathcal{B}$ can be characterized as follows: $f \in \mathcal{B}$ if and only if $f \in \mathcal{A}$ or $\mathcal{N}(f)$ is non-degenerate and has only finitely many segments with non-positive slope. The following lemma and its proof will be very useful in Section 5.

LEMMA 3. *If $f(z, X) \in \mathcal{B}$, then there exists $c \in \mathbb{Z}$ such that $f(z, z^c X) \in \mathcal{A}$.*

PROOF. Let $f = \sum_{k=0}^{+\infty} f_k(z) X^k \in \mathcal{B}$ and, for $k \in \mathbb{N}$, $v_k = v_z(f_k)$. We then have:

$$\begin{aligned} f &= X^\nu \left(z^{v_0} \tilde{f}_0 + z^{v_1} \tilde{f}_1 X + z^{v_2} \tilde{f}_2 X^2 + \dots \right), \\ &= X^\nu z^{v_0} \left(\tilde{f}_0 + z^{v_1 - c - v_0} \tilde{f}_1 \tilde{X} + z^{v_2 - 2c - v_0} \tilde{f}_2 \tilde{X}^2 + \dots \right), \end{aligned}$$

where $\nu \geq 0$, $\tilde{f}_0 \neq 0$, $\tilde{X} = z^c X$, and for $k \in \mathbb{N}$, $v_z(\tilde{f}_k) = 0$. It is thus sufficient to prove that there exists $c \in \mathbb{Z}$ so that $v_k \geq ck + v_0$, for all $k \in \mathbb{N}$. Using (6), there exists $\alpha, \beta \in \mathbb{Q}$ such that $v_k \geq \alpha k + \beta$, for $k \in \mathbb{N}$. A straightforward calculation shows that if we take $c = \lfloor \alpha \rfloor$ if $\beta \geq v_0$, and $c = \lfloor \alpha + \beta - v_0 \rfloor$ if $\beta < v_0$, where $\lfloor a \rfloor \in \mathbb{Z}$ denotes the integer part of $a \in \mathbb{Q}$, then we have $v_k \geq \alpha k + \beta \geq ck + v_0$ which ends the proof. $\square$

For $f \in \mathcal{B}$, let us define α_f to be the greatest α that we can take in (6). If $f \in \mathcal{A}$, then $\alpha_f = 0$ otherwise it corresponds to the greatest non-positive slope of the segments of $\mathcal{N}(f)$. Then, for $f \in \mathcal{B}$, we can define:

$$w(f) = \min_{k \in \mathbb{N}} (v_z(f_k) - \alpha_f k).$$

Note that geometrically, $w(f)$ corresponds to the y -coordinate of the point where the line of the greatest slope of $\mathcal{N}(f)$ crosses the y -axis. Moreover, one can check that, for all $f, g \in \mathcal{B}$,

$$w(f + g) \geq \min(w(f), w(g)), \quad w(fg) \geq w(f) + w(g). \quad (7)$$

For more details concerning Newton polygons in this setting, we refer to [16, 17, 15].

LEMMA 4. *Let $\mathcal{F}$ denote the quotient field of $\mathcal{B}$. Then, we have $\inf_{f \in \mathcal{F}, f \neq 0} w(f^{-1} f') = -q$.*

PROOF. Let $f \in \mathcal{F}$, $f \neq 0$. Using the characterization (6), we can check that f can be written as $f = X^\nu \tilde{f}$ with $\nu \in \mathbb{Z}$ and $\tilde{f} \in \mathcal{B}$. We thus obtain $f' = X^\nu (\nu z^{-q} a(z) \tilde{f} + \tilde{f}')$ so that $f^{-1} f' = \nu z^{-q} a(z) + \tilde{f}^{-1} \tilde{f}'$. Consequently, from (7), we have $w(f^{-1} f') \geq \min(-q, w(\tilde{f}^{-1} \tilde{f}')) \geq -q$ and the minimum is reached for $f = X$. $\square$

We refer to [8, 4] for more explanations on the quantity computed in Lemma 4.

REMARK 1. *In our algorithms, the series of the form $f = \sum_{k=0}^{+\infty} f_k(z) X^k$, with $f_k(z) \in \mathbb{C}[[z]]$, are represented by their truncation up to a certain order with respect to X and z . In the sequel, the sentence “we can compute f ” means that we are able to compute its truncation up to an arbitrary order.*

2.2 Meromorphic linear differential systems

In this subsection, we first recall some definitions concerning formal meromorphic linear differential systems of the form (2). Then, we review the algorithm of [3] for computing a formal fundamental matrix of solutions of the form (3). In the sequel, to simplify, we will often use the notation $[A]$ for a linear differential system $Y' = AY$. Therefore System (2) is denoted by $[z^{-p} A]$. The integer $p - 1 \geq 0$ is then called the *Poincaré rank* of $[z^{-p} A]$.

If we perform a change of variables $Y = TZ$ with $T \in \text{GL}_n(\mathbb{C}((z)))$ in a system $[A]$, then the vector in terms of new variables Z satisfies $Z' = T^{-1}(AT - T')Z$. If F denotes an algebraic field extension of $\mathbb{C}((z))$, then two formal meromorphic linear differential systems $[A]$ and $[B]$ are *equivalent over F* if there exists an invertible formal meromorphic matrix transformation $T \in \text{GL}_n(F)$ such that $B = T[A] := T^{-1}(AT - T')$. Note that, we have $T[z^{-p} A] = z^{-p} T^{-1}(AT - z^p T')$ so that two differential systems $[z^{-p} A]$ and $[z^{-p} B]$ are equivalent over F if there exists $T \in \text{GL}_n(F)$ such that $B = T^{-1}(AT - z^p T')$. Moreover if $B = T[A]$ and $Y_{[B]}$ is a formal fundamental matrix of solutions of $[B]$, then

$Y_{[A]} = T Y_{[B]}$ is a formal fundamental matrix of solutions of $[A]$.

It is known (see, for example, [19, 21, 3]), that a formal fundamental matrix of solutions of (2) has the form (3). An algorithm to compute such a formal fundamental matrix of solutions is developed in [3]. The simplest case appears when $p \leq 1$ since in this case we have $r = 1$, $Q = 0$ and the matrices Φ and Λ can be computed by the algorithm given in [5]. When $p > 0$, then using Moser’s algorithm ([18, 3]) we can reduce the system (2) to an equivalent one with minimal Poincaré rank $\tilde{p} - 1$. If $\tilde{p} = 1$, then $z = 0$ is a *regular singularity* and we can still apply the algorithm of [5] to get a formal fundamental matrix of solutions with $r = 1$ and $Q = 0$. If the minimal Poincaré rank that we can get via meromorphic transformations is not equal to 0, then we have an *irregular singularity* and we shall proceed to compute a formal fundamental matrix of solutions with an exponential part $Q \neq 0$. The technique of [3] consists in reducing the system (2) to several independent systems each of which is either a system of the form (2) with $p = 1$ or a first order scalar differential equation ($n = 1$). This is achieved as follows:

1. We apply a first transformation to split the system into a set of differential systems of the same form where each system has a leading matrix $A(0)$ with only one eigenvalue and we now consider one by one the resulting systems (of smaller size).
2. A second transformation is performed to get a new system with a nilpotent leading matrix (i.e., we shift the non-zero eigenvalues of $A(0)$ to 0) and, if its Poincaré rank is not minimal, we apply Moser’s algorithm ([18, 3]) to get an equivalent system with minimal Poincaré rank.
3. If the resulting system still has a Poincaré rank strictly greater than 0 and $A(0)$ is nilpotent, then we compute its Katz’ invariant κ ([14, 3]) and we perform the ramification $z = t^m$ where m is the smallest positive integer such that $m\kappa$ is an integer. Applying one more time Moser’s algorithm ([18, 3]) yields a new system with Poincaré rank $m\kappa$ and $A(0)$ not nilpotent. Then, we can apply another transformation to split the system into several systems of smaller size.
4. We apply recursion.

At each step of this process, we either reduce the Poincaré rank or the size of the system so that the process ends with $p = 1$ or $n = 1$. For more details, see [3].

REMARK 2. *In the algorithm sketched above, the discussion is mainly based on the comparison between the quantities p and 1 or $-p$ and -1 . Note that -1 corresponds to $\inf_{f \in \mathbb{C}((z)), f \neq 0} v_z(f^{-1} f')$ (see [4]). This justifies the fact that in the sequel, when we consider System (1), we compare $-p$ and $-q$ to distinguish the different cases (see Lemma 4).*

3. MAIN RESULTS

In the present paper, we study systems of the form

$$Y' = \tilde{A}(z, X) Y, \quad (8)$$

where $' = d/dz$, X is a transcendental element defined by (1b) and the square matrix $\tilde{A}(z, X)$ is assumed to have entries in the ring $\mathcal{A}$ defined by (5). As a consequence, we can

write $\tilde{A}(z, X) = \sum_{k=0}^{+\infty} \tilde{A}_k(z) X^k = z^{-p} A(z, X)$, with

$$A(z, X) = \sum_{k=0}^{+\infty} A_k(z) X^k, \quad A_k(z) \in \mathbb{M}_n(\mathbb{C}[[z]]). \quad (9)$$

In the sequel, we further assume that $A(0, 0) = A_0(0) \neq 0$ and $p \in \mathbb{N}^*$, so that we consider systems of the form (1) denoted by $\mathcal{S}_{A,a}^{p,q}$, namely,

$$\mathcal{S}_{A,a}^{p,q} : \begin{cases} Y' = z^{-p} A(z, X) Y, & p \in \mathbb{N}^*, A(0, 0) \neq 0, \\ X' = z^{-q} a(z) X, & q \geq 2, a(z) \in \mathbb{C}[[z]], a(0) \neq 0, \end{cases}$$

where $A(z, X)$ is given by (9) with $A_0(0) \neq 0$. If we naturally extend the valuation v defined in Subsection 2.1 to matrices $A = (a_{i,j}) \in \mathbb{M}_n(\mathcal{A})$ by $v(A) = \min_{i,j} v(a_{i,j})$ (and we extend similarly the z -adic valuation to matrices), then the assumption $A_0(0) \neq 0$ means that

$$-p = v(\tilde{A}(z, X)) = \min_{k \in \mathbb{N}} v_z(\tilde{A}_k(z)) = v_z(\tilde{A}_0(z)).$$

The contribution of this paper consists in investigating the effective computation of a formal fundamental matrix of solutions of $\mathcal{S}_{A,a}^{p,q}$ of the form (4). See also Section 5 for extensions to more general classes of differential systems, e.g., systems with coefficients in the ring $\mathcal{B}$ defined by (6).

By analogy to the terminology used in the study of formal meromorphic linear differential systems, the systems of the form (1) can be called X -regular in the sense that there is no exponential part depending on X in a formal fundamental matrix of solutions. Note that this *irregular* part in X will appear when we consider systems of the form (8) but where instead of choosing the entries of $\tilde{A}(z, X)$ in $\mathcal{A}$ defined by (5), we allow terms with negative orders in X . Considering systems with poles in X is the subject of a work in progress.

The following of the paper is devoted to the construction of an algorithm for computing a formal fundamental matrix of solutions (4) of $\mathcal{S}_{A,a}^{p,q}$. We distinguish different cases by comparing the values of p and q (see Remark 2) and we shall prove that:

1. If $p < q$, $\mathcal{S}_{A,a}^{p,q}$ can be reduced to the formal meromorphic linear differential system $[z^{-p} A_0]$;
2. If $p = q$, $\mathcal{S}_{A,a}^{p,q}$ can first be reduced to another system of the same form $\mathcal{S}_{\tilde{A},a}^{\tilde{p},q}$ satisfying that either $\tilde{p} < q$ or $\tilde{p} = q$ and $\tilde{A}_0(0)$ does not have eigenvalues that differ by an integer multiple of $a(0)$. Moreover, in the latter case $\mathcal{S}_{\tilde{A},a}^{\tilde{p},q}$ can be reduced to the formal meromorphic linear differential system $[z^{-p} \tilde{A}_0]$;
3. If $p > q$, we can adapt the process of [3] reviewed in Subsection 2.2 to reduce $\mathcal{S}_{A,a}^{p,q}$ to several independent systems of the same form each of which either satisfies $p \leq q$ or $n = 1$ (scalar differential equation).

4. ALGORITHM

4.1 The case $p \leq q$

DEFINITION 1. We say that a system $\mathcal{S}_{A,a}^{p,q}$ satisfies the property (P) if either $p < q$ or $p = q$ and $A_0(0)$ has no eigenvalues that differ by an integer multiple of $a(0)$.

The following lemma will be used in the proof of Theorem 1.

LEMMA 5. Let m be an integer such that $m \geq 2$ and $M \in \mathbb{M}_r(\mathbb{C}[[z]])$, $N \in \mathbb{M}_s(\mathbb{C}[[z]])$ be two square matrices. If $M(0)$ and $N(0)$ have no common eigenvalues, then, for every $V \in \mathbb{M}_{r \times s}(\mathbb{C}[[z]])$, there exists a unique matrix $U \in \mathbb{M}_{r \times s}(\mathbb{C}[[z]])$ such that

$$z^m U' = M U - U N - V. \quad (10)$$

PROOF. Let $M = \sum_{k=0}^{+\infty} M_k z^k$, $N = \sum_{k=0}^{+\infty} N_k z^k$, and $V = \sum_{k=0}^{+\infty} V_k z^k$. Plugging the ansatz $U = \sum_{k=0}^{+\infty} U_k z^k$ into (10) and identifying the coefficients in z using the fact that $m \geq 2$, we get: $M(0) U_0 - U_0 N(0) = V_0$, and, for $k \geq 1$,

$$M(0) U_k - U_k N(0) = (k - m + 1) U_{k-m+1} - \sum_{i=1}^k (M_i U_{k-i} - U_{k-i} N_i) + V_k,$$

with the convention $U_{-k} = 0$ for all $k > 0$. We conclude using a well known result (see for example [12, Corollary S2.3, p. 347] or [3, Remark 6]) asserting that if $M \in \mathbb{M}_r(\mathbb{C})$ and $N \in \mathbb{M}_s(\mathbb{C})$ have no common eigenvalues, then, for every matrix V in $\mathbb{M}_{r \times s}(\mathbb{C})$, we can compute a unique matrix U in $\mathbb{M}_{r \times s}(\mathbb{C})$ such that $M U - U N = V$. $\square$

THEOREM 1. Let us consider a system $\mathcal{S}_{A,a}^{p,q}$ satisfying the property (P). Then, $\mathcal{S}_{A,a}^{p,q}$ can be reduced to the formal meromorphic linear differential system $[z^{-p} A_0(z)]$. More precisely, we can compute an invertible matrix transformation

$$T = I_n + T_1(z) X + T_2(z) X^2 + \cdots, \quad T_k(z) \in \mathbb{M}_n(\mathbb{C}[[z]]),$$

where I_n denotes the identity matrix of size n , such that $A_0 = T^{-1}(A T - z^p T')$, i.e., $z^p T' = A T - T A_0$.

PROOF. If we take an ansatz $T = \sum_{k=0}^{+\infty} T_k(z) X^k$, we have:

$$z^p T' = \sum_{k=0}^{+\infty} (z^p T'_k(z) + k z^{p-q} a(z) T_k(z)) X^k,$$

and

$$A T - T A_0 = \sum_{k=0}^{+\infty} \left(\sum_{m=0}^k (A_m T_{k-m}) - T_k A_0 \right) X^k.$$

By identification, we then obtain:

$$z^p T'_0 = A_0 T_0 - T_0 A_0, \quad (11)$$

and, for $k \geq 1$,

$$z^p T'_k = A_0 T_k - T_k (A_0 + k z^{p-q} a(z) I_n) + \sum_{m=1}^k A_m T_{k-m}. \quad (12)$$

Multiplying (12) by z^{q-p} , we get:

$$z^q T'_k = (z^{q-p} A_0) T_k - T_k (z^{q-p} A_0 + k a(z) I_n) + z^{q-p} \sum_{m=1}^k A_m T_{k-m}. \quad (13)$$

The matrix $T_0 = I_n$ is a trivial solution of (11) and (13) can be solved using Lemma 5 below. Indeed, for all $k \geq 1$, (13) is of the form (10). Now if $p < q$, it satisfies the hypotheses of Lemma 5 since we have $q \geq 2$, $q - p > 0$ and $a(0) \neq 0$. Moreover, if $p = q$, the hypotheses of Lemma 5 are also fulfilled since $q \geq 2$ and $A_0(0)$ has no eigenvalues that differ by an integer multiple of $a(0)$. $\square$

The proofs of Theorem 1 and Lemma 5 provide an algorithm for computing a matrix $T = I_n + T_1(z)X + T_2(z)X^2 + \dots$ such that $z^p T' = AT - TA_0$. Choosing the first coefficient $T_0 = I_n$ ensures that the computed matrix transformation T is invertible. Moreover, from Lemma 5, we obtain unique matrices $T_k(z)$ the entries of which are formal power series, i.e., $T_k(z) \in \mathbb{M}_n(\mathbb{C}[[z]])$.

Note that when $p < q$, Theorem 1 can be applied even if $A_0(z) = 0$ in which case the matrix transformation T yields a formal fundamental matrix of solutions.

To conclude for the case $p \leq q$, it remains to consider the case when the system $\mathcal{S}_{A,a}^{p,q}$ does not satisfy the property (P), i.e., $p = q$ and $A_0(0)$ has eigenvalues that differ by an integer multiple of $a(0)$. We have the following result:

PROPOSITION 1. *A system $\mathcal{S}_{A,a}^{p,q}$ such that $p = q$ and $A_0(0)$ has eigenvalues that differ by an integer multiple of $a(0)$ can be reduced to a system $\mathcal{S}_{A,a}^{\tilde{p},q}$ satisfying the property (P).*

PROOF. We adapt the process used in [21, Section 17] in a similar situation to our case. First, we arrange the eigenvalues of $A_0(0) \neq 0$ into disjoint sets, so that the elements of each set differ by an integer multiple of $a(0)$. Let $\lambda_1, \dots, \lambda_s$ be the elements of such a set and sort them such that if $k_i = a(0)^{-1}(\lambda_i - \lambda_s)$, $i = 1, \dots, s-1$, then we have $k_1 > \dots > k_{s-1} > 0$. Let $\lambda_{s+1}, \dots, \lambda_r$ denote the other eigenvalues of $A_0(0)$ and for $i = 1, \dots, r$, let m_i be the multiplicity of λ_i . After applying an adequate constant matrix transformation, we may assume without loss of generality that the matrix $A_0(0)$ has the following diagonal form: $A_0(0) = \text{diag}(A_0^{11}(0), A_0^{22}(0))$, and the only eigenvalue of the matrix $A_0^{11}(0)$ is λ_1 . Now, since $p = q \geq 2$, and $A_0(0)$ has more than two distinct eigenvalues, the formal meromorphic linear differential system $[z^{-p} A_0]$ is irregular so that there exists a matrix transformation

$$P = I_n + P_1 z + P_2 z^2 + \dots \in \mathbb{M}_n(\mathbb{C}[[z]]), \quad (14)$$

that splits $[z^{-p} A_0]$ (*Splitting lemma*: see [21, Theorem 11.1, p. 54] or [3, Theorem 2]). Note that the condition $P(0) = I_n$ guarantees the fact that its inverse P^{-1} has formal power series entries so that $P^{-1} A_k P \in \mathbb{M}_n(\mathbb{C}[[z]])$ holds for all $k \in \mathbb{N}$. After applying such a *splitting transformation* P , we may assume that $A_0(z)$ has the diagonal form $A_0(z) = \text{diag}(A_0^{11}(z), A_0^{22}(z))$, where $A_0^{11}(z)$ is an $m_1 \times m_1$ matrix and $A_0^{11}(0)$ has only one eigenvalue λ_1 . Now, applying the transformation

$$U = \text{diag}(X I_{m_1}, I_{n-m_1}), \quad (15)$$

one can check that we get a new equivalent system $\mathcal{S}_{B,a}^{p,q}$ with

$$Y' = z^{-p} \underbrace{(B_0(z) + B_1(z)X + \dots)}_{B(z,X)} Y,$$

where $B(z, X) = U^{-1}(A(z, X)U - z^p U')$. Now, if we partition $A_1(z) \in \mathbb{M}_n(\mathbb{C}[[z]])$ accordingly, i.e.,

$$A_1(z) = \begin{pmatrix} A_1^{11}(z) & A_1^{12}(z) \\ A_1^{21}(z) & A_1^{22}(z) \end{pmatrix}, \quad A_1^{11}(z) \in \mathbb{M}_{m_1}(\mathbb{C}[[z]]),$$

then, a straightforward calculation shows that:

$$B_0(z) = \begin{pmatrix} A_0^{11}(z) & A_1^{12}(z) \\ 0 & A_0^{22}(z) \end{pmatrix} - a(z) \begin{pmatrix} I_{m_1} & 0 \\ 0 & 0 \end{pmatrix}.$$

The eigenvalues of $B_0(0)$ are thus $\lambda_1 - a(0), \lambda_2, \dots, \lambda_r$. Thus, after $k_1 + \dots + k_{s-1}$ steps, the eigenvalues $\lambda_1, \dots, \lambda_s$ are

reduced to the single eigenvalue λ_s which has multiplicity $m_1 + \dots + m_s$. Finally, by applying the same process to the other sets of eigenvalues that differ by an integer multiple of $a(0)$, one obtains a new system $\mathcal{S}_{A,a}^{\tilde{p},q}$ which satisfies the property (P). $\square$

The proof of the previous proposition is constructive and provides an algorithm to compute a matrix transformation T such that $\tilde{A} = T^{-1}(AT - z^p T')$ with the notations of Proposition 1. Moreover we get a matrix $T \in \mathbb{M}_n(\mathbb{C}[[z]] [[X]])$ which is invertible since it is a product of constant matrices, matrices of the form (14) and matrices of the form (15).

REMARK 3. *During the process described in the proof of Proposition 1, the leading matrix $A_0(0)$ can at some stage be reduced to the zero matrix. In this case, we apply a normalization of the form $X \rightarrow z^m X$, $m \in \mathbb{Z}^*$ as explained in Section 5 to get a new system satisfying $p < q$.*

REMARK 4. *Analyzing the proof of Proposition 1, we can see that for each subset of eigenvalues of $A_0(0)$ that differ by an integer multiple of $a(0)$, the number of terms needed in the X -adic expansion of the matrix $A(z, X)$ is exactly $k_1 + \dots + k_{s-1} = a(0)^{-1}(\lambda_1 - \lambda_s)$.*

We finally get the following consequence of Theorem 1 and Proposition 1:

THEOREM 2. *Let $\mathcal{S}_{A,a}^{p,q}$ be such that $p \leq q$. Then, using the above results, we can compute a formal fundamental matrix of solutions of $\mathcal{S}_{A,a}^{p,q}$ of the form*

$$Y = \left(\sum_{k=0}^{+\infty} T_k(z) X^k \right) \Phi(t) t^\Lambda \exp(Q(1/t)),$$

with the notation of (3) and $T_k(z) \in \mathbb{M}_n(\mathbb{C}[[z]])$.

EXAMPLE 1. *Let us consider the Schrödinger equation with Yukawa potential given by:*

$$-\frac{\hbar^2}{2m} \frac{d^2 y}{dr^2} - g^2 \frac{e^{-k m r}}{r} y = E y,$$

where $\hbar, m, g$ and k are some physical constants and E is the energy eigenvalue (see [13]). This is a second order linear differential equation over $\mathbb{C}$ with an essential singularity at the point at infinity. The associated linear differential system is:

$$\frac{dY}{dr} = r^{-1} A(r) Y, \quad A(r) = \begin{pmatrix} 0 & r \\ -\frac{2mg^2}{\hbar^2} e^{-k m r} & -\frac{2mE}{\hbar^2} r \end{pmatrix}.$$

To compute the formal solutions at infinity, we perform the change of variable $z = 1/r$. Setting $X = e^{-\frac{k m}{z}}$, we then get the system:

$$Y' = z^{-2} A(z, X) Y, \quad X' = z^{-2} k m X, \quad (16)$$

where

$$A(z, X) = \begin{pmatrix} 0 & -1 \\ \frac{2mg^2}{\hbar^2} z X + \frac{2mE}{\hbar^2} & 0 \end{pmatrix}.$$

We thus have a system $\mathcal{S}_{A,a}^{2,2}$ with $a(z) = a(0) = k m$ and

$$A_0(z) = A_0(0) = \begin{pmatrix} 0 & -1 \\ \frac{2mE}{\hbar^2} & 0 \end{pmatrix}.$$

The eigenvalues of $A_0(0)$ are $\pm i\frac{\sqrt{2mE}}{\hbar}$ and the difference between them is clearly not an integer multiple of km so that (16) satisfies the property (P). Then, Theorem 1 guarantees the existence of a formal matrix transformation $T = \sum_{k=0}^{+\infty} T_k(z) X^k$ with $T_0(z) = I_2$ and $T_k(z) \in \mathbb{M}_2(\mathbb{C}[[z]])$ such that we have $z^2 T' = AT - TA_0$. The matrix T can be computed (up to arbitrary orders - see Remark 1) by solving Equations (13) using Lemma 5. Note that the meromorphic system $[z^{-2} A_0]$ is the free particle Schrödinger equation at $+\infty$. A formal fundamental matrix of solutions of $[z^{-2} A_0]$ is given by

$$e^{\int z^{-2} A_0 dz} = \begin{pmatrix} \cos\left(\frac{\sqrt{2mE}}{\hbar z}\right) & \frac{\hbar}{\sqrt{2mE}} \sin\left(\frac{\sqrt{2mE}}{\hbar z}\right) \\ -\frac{\sqrt{2mE}}{\hbar} \sin\left(\frac{\sqrt{2mE}}{\hbar z}\right) & \cos\left(\frac{\sqrt{2mE}}{\hbar z}\right) \end{pmatrix},$$

and we get that the form of a formal fundamental matrix of solutions of (16) is $(\sum_{k=0}^{+\infty} T_k(z) X^k) e^{\int z^{-2} A_0 dz}$.

4.2 The case $p > q$

Here, we shall adapt to our setting the strategy of the algorithm of [3] reviewed in Subsection 2.2 to study formal meromorphic linear differential systems at an irregular singularity. It consists in applying a (finite) sequence of matrix transformations and changes of variables in order to reduce to the case of a system $\mathcal{S}_{A,a}^{p,q}$ where either $\tilde{p} \leq q$ (Subsection 4.1) or $n = 1$ (scalar linear differential equations).

The scalar case can be handled as follows. Let $Y_0(z) = \exp(\int z^{-p} A_0(z) dz)$ be the solution of the leading equation $[z^{-p} A_0(z)]$. Performing the change of variables $Y = Y_0 Z$ and using the fact that $Y_0(z)^{-1}$ satisfies $[-z^{-p} A_0(z)]$, we obtain that the new variable Z satisfies the differential equation $Z' = z^{-p} (A_1(z) X + A_2(z) X^2 + \dots) Z$. Using a normalization of the form $X \rightarrow z^c X$ with $c \in \mathbb{Z}$, we are then reduced to a new differential equation where $p < q$ and $A_0(z) = 0$ (see Remark 6 in Section 5 below). Consequently, we can find a transformation $T \in \mathbb{C}[[z]][[X]]$ such that the equation is reduced to [0]. The fundamental solution is thus $Y(z, X) = Y_0(z) T(z, z^c X)$ which is of the form (4).

We first show how to split systems having a leading matrix $A_0(0)$ with at least two distinct eigenvalues.

PROPOSITION 2. *Let $\mathcal{S}_{A,a}^{p,q}$ be such that $p > q$ and assume that $A_0(0)$ has $r \geq 2$ distinct eigenvalues. Then, $\mathcal{S}_{A,a}^{p,q}$ can be split into r systems $\mathcal{S}_{A^{[i]},a}^{p,q}$, where each $A_0^{[i]}(0)$ has only one eigenvalue. More precisely, we can compute a matrix transformation*

$$T = I_n + T_1(z) X + T_2(z) X^2 + \dots, \quad T_k(z) \in \mathbb{M}_n(\mathbb{C}[[z]]),$$

such that

$$z^p T' = A(z, X) T - T \operatorname{diag}(A^{[1]}(z, X), \dots, A^{[r]}(z, X)),$$

and each $A^{[i]}(0, 0)$ has only one eigenvalue.

PROOF. For simplicity, we restrict to the case $r = 2$, which will be used to prove Theorem 3 below. We mimic the strategy of [21, Chapter 4, Section 11]. Applying the splitting lemma (see [21, Theorem 11.1, p. 54] or [3, Theorem 2]) as in the proof of Proposition 1, we can first assume that $A_0(z) = \operatorname{diag}(A_0^{11}(z), A_0^{22}(z))$ and $A_0^{11}(0)$ and $A_0^{22}(0)$ have no common eigenvalues. We are going to prove that we can compute a matrix transformation T of the desired form such that we have $z^p T' = A(z, X) T - T \operatorname{diag}(B^{[1]}(z, X), B^{[2]}(z, X))$,

where $B^{[i]}(0, 0) = A_0^{ii}(0)$ for $i = 1, 2$. Plugging an ansatz $T = \sum_{k=0}^{+\infty} T_k(z) X^k$, where

$$T_0(z) = I_n, \quad T_k(z) = \begin{pmatrix} 0 & T_k^{12}(z) \\ T_k^{21}(z) & 0 \end{pmatrix}, \quad \forall k \in \mathbb{N}^*,$$

into the relation $z^p T' = A(z, X) T - T B(z, X)$, we get

$$z^p T'_0 = A_0 T_0 - T_0 B_0, \quad (17)$$

and, for $k \geq 1$,

$$z^p T'_k = A_0 T_k - T_k (B_0 + k z^{p-q} a(z) I_n) + \sum_{m=1}^{k-1} (A_m T_{k-m} - T_{k-m} B_m) + A_k - B_k. \quad (18)$$

Now, $T_0 = I_n$ and $B_0(z) = A_0(z)$ provides a trivial solution of (17). If we take the corresponding partition by blocks

$$\sum_{m=1}^{k-1} (A_m T_{k-m} - T_{k-m} B_m) + A_k = \begin{pmatrix} W_k^{11}(z) & W_k^{12}(z) \\ W_k^{21}(z) & W_k^{22}(z) \end{pmatrix},$$

and if we set $B_k(z) = \operatorname{diag}(W_k^{11}(z), W_k^{22}(z))$, then Equation (18) reduces to

$$\begin{cases} z^p (T_k^{12})' = A_0^{11} T_k^{12} - T_k^{12} (A_0^{22} + k z^{p-q} a(z) I) + W_k^{12}, \\ z^p (T_k^{21})' = A_0^{22} T_k^{21} - T_k^{21} (A_0^{11} + k z^{p-q} a(z) I) + W_k^{21}. \end{cases}$$

Finally, using in particular the fact that $p > q \geq 2$, the hypotheses of Lemma 5 are satisfied. This implies that both equations can be solved for matrices T_k^{12} and T_k^{21} with entries in $\mathbb{C}[[z]]$ which ends the proof. $\square$

To compute a formal fundamental matrix of solutions of $\mathcal{S}_{A,a}^{p,q}$, we shall need to perform ramifications of the variable z . The following straightforward lemma shows the effect of such a ramification on a system $\mathcal{S}_{A,a}^{p,q}$.

LEMMA 6. *Let $r \in \mathbb{N}^*$ be a positive integer. Then, performing the ramification $z = t^r$ in the differential system $\mathcal{S}_{A,a}^{p,q}$ in the variable z yields a new differential system $\mathcal{S}_{\tilde{A},\tilde{a}}^{p,\tilde{q}}$ in the variable t , where we have $\tilde{p} = r(p-1)+1$, $\tilde{q} = r(q-1)+1$, and $\tilde{A}(t, X) = r A(t^r, X)$, $\tilde{a}(t) = r a(t^r)$.*

Note that, from Lemma 6, performing a ramification on $\mathcal{S}_{A,a}^{p,q}$ does not affect the sign of the quantity $p - q$.

THEOREM 3. *Let $\mathcal{S}_{A,a}^{p,q}$ be such that $p > q$. Then, we can compute a formal fundamental matrix of solutions of $\mathcal{S}_{A,a}^{p,q}$ of the form*

$$Y = \left(\sum_{k=0}^{+\infty} \Phi_k(t) X^k \right) t^\Lambda \exp(Q(1/t)),$$

with the notation of (3) and $\Phi_k(t) \in \mathbb{M}_n(\mathbb{C}((t)))$.

PROOF. Let us consider the leading formal meromorphic linear differential system $[z^{-p} A_0(z)]$ and assume that the ramification appearing in its formal fundamental matrix of solutions of the form (3) is $z = t^r$. We first perform the ramification $z = t^r$ in the whole system $\mathcal{S}_{A,a}^{p,q}$ so that we get a new system $\mathcal{S}_{\tilde{A},\tilde{a}}^{\tilde{p},\tilde{q}}$ satisfying the relations of Lemma 6. To simplify the notation in the sequel, we still denote by $\mathcal{S}_{A,a}^{p,q}$ this ramified system and by z the variable. From the theory of formal meromorphic linear differential systems, we

can compute a matrix transformation $T \in \text{GL}_n(\mathbb{C}((z)))$ such that we have

$$T[z^{-p} A_0(z)] = z^{-p} \text{diag}(B_1(z), \dots, B_s(z)),$$

where, for $i = 1, \dots, s$, $B_i(z) = \omega_i(z) I_{n_i} + z^{p-1} \Lambda_i$, with $\omega_i(z) \in \mathbb{C}[z]$ of degree $\leq (p-2)$ and $\Lambda_i \in \mathbb{M}_{n_i}(\mathbb{C})$ (see [21, 3]). Applying this matrix transformation T to the whole differential system $\mathcal{S}_{A,a}^{p,q}$, we can assume without loss of generality that in $\mathcal{S}_{A,a}^{p,q}$, we have $A_0(z) = \text{diag}(B_1(z), \dots, B_s(z))$. Then, we apply the transformation

$$T = \exp\left(\int z^{-p} \omega_1(z) dz\right) I_n \in \text{GL}_n(\mathbb{C}((z))),$$

to $\mathcal{S}_{A,a}^{p,q}$. This provides a new differential system $\mathcal{S}_{\tilde{A},a}^{p,q}$ with

$$\tilde{A}_0(z) = A_0(z) - \omega_1(z) I_n = \text{diag}(\tilde{B}_1(z), \dots, \tilde{B}_s(z)),$$

with $\tilde{B}_1(z) = z^{p-1} \Lambda_1$ and

$$\tilde{B}_i(z) = \underbrace{(\omega_i(z) - \omega_1(z))}_{\tilde{\omega}_i(z)} I_{n_i} + z^{p-1} \Lambda_i, \quad i = 2, \dots, s.$$

Let us now define the following positive integer:

$$k = \min_{i=2, \dots, s} v_z(\tilde{\omega}_i(z)) \leq p-2.$$

Up to a permutation (that we apply to the whole differential system $\mathcal{S}_{\tilde{A},a}^{p,q}$), we can then reorder the blocks of the matrix $\tilde{A}_0(z)$ such that $\tilde{A}_0(z) = \text{diag}(\tilde{A}_0^{11}(z), \tilde{A}_0^{22}(z))$, with

$$\tilde{A}_0^{11}(z) = \text{diag}(\{\tilde{B}_j(z)\}_{j \in J_1}), \quad \forall j \in J_1, v_z(\tilde{B}_j(z)) > k,$$

$$\tilde{A}_0^{22}(z) = \text{diag}(\{\tilde{B}_j(z)\}_{j \in J_2}), \quad \forall j \in J_2, v_z(\tilde{B}_j(z)) = k,$$

so that $v_z(\tilde{A}_0^{11}(z)) > k$ and $v_z(\tilde{A}_0^{22}(z)) = k$. We now apply the normalization $X \rightarrow z^{-k} X$ which yields a new differential system $\mathcal{S}_{\tilde{A},a}^{\bar{p},q}$ satisfying the condition $\bar{p} = p - k$ and $\tilde{A}_0(z) = z^{-k} \tilde{A}_0(z)$ (see Section 5). If $p - k \leq q$, then we are reduced to the case of Subsection 4.1. Otherwise we are reduced to a system $\mathcal{S}_{\tilde{A},a}^{\bar{p},q}$ satisfying both $\bar{p} > q$ and

$$\tilde{A}_0(z) = \text{diag}(\tilde{A}_0^{11}(z), \tilde{A}_0^{22}(z)), \quad \tilde{A}_0^{ii}(z) = z^{-k} \tilde{A}_0^{ii}(z), \quad i = 1, 2.$$

Consequently, $\tilde{A}_0^{11}(0) = 0$ has the sole eigenvalue 0 and $\tilde{A}_0^{22}(0)$ has only non-zero eigenvalues so that we can apply Proposition 2 for $r = 2$ to split the whole differential $\mathcal{S}_{\tilde{A},a}^{\bar{p},q}$ into two uncoupled differential systems. Iterating this process on each system obtained yields the desired result. Indeed, we will necessarily end up with a differential system satisfying either $p \leq q$ (see Subsection 4.1) or $n = 1$ (scalar case). $\square$

In our implementation, instead of proceeding as in the proof of Theorem 3 above, we rather follow the distinct steps of the algorithm of [3] applied to the leading formal meromorphic linear differential system $[z^{-p} A_0(z)]$. Namely, we apply one after the other the steps reviewed in Subsection 2.2 to $[z^{-p} A_0(z)]$ and at each step we perform the transformations needed (e.g., splitting, shift, Moser's reduction, ramification, ...) to the whole system $\mathcal{S}_{A,a}^{p,q}$.

REMARK 5. The form (4) of a formal fundamental matrix of solutions of $\mathcal{S}_{A,a}^{p,q}$ could be confusing as one may think that no negative power in X is involved. It is not true as, for example, for all $\lambda \in \mathbb{C}^*$ and X defined by (1b), the

scalar equation $Y' = \lambda z^{-q} a(z) Y$ admits the solution $Y = \exp(\int \lambda z^{-q} a(z) dz)$ which is exactly X^λ . Such a negative power of X can indeed be hidden in the part $t^\Lambda \exp(Q(1/t))$.

5. WIDER CLASSES OF SYSTEMS

In this section, we show that the algorithm of Section 4 can handle a wider class of systems than $\mathcal{S}_{A,a}^{p,q}$. Consider a differential system of the form

$$\begin{cases} Y' = z^{-p} A(z, X) Y, & p \in \mathbb{N}^*, A(z, X) \neq 0, \\ X' = z^{-q} a(z) X, & q \geq 2, a(z) \in \mathbb{C}[[z]], a(0) \neq 0, \end{cases} \quad (19)$$

where $A(z, X) = \sum_{k=0}^{+\infty} A_k(z) X^k$, $A_k(z) \in \mathbb{M}_n(\mathbb{C}((z)))$.

The algorithm presented in Section 4 applies to an input system $\mathcal{S}_{A,a}^{p,q}$ which is supposed to have its coefficients living in the ring $\mathcal{A}$ defined by (5) and to satisfy $A_0(0) \neq 0$. However during the reduction process described in Section 4 one naturally may obtain systems with coefficients that no longer belong to $\mathcal{A}$ (i.e., $v_z(A_k) \rightarrow -\infty$, when $k \rightarrow +\infty$). Moreover one can also be confronted to systems of the form (19) with $v_z(A_0) \neq 0$ (in particular when $A_0(0) = 0$). We show here that we can always overcome these difficulties by using *normalizations* of the form $X \rightarrow z^m X$ and allowing the coefficients of the input system to live in the ring $\mathcal{B} \supsetneq \mathcal{A}$ defined by (6). We first give the following straightforward lemma:

LEMMA 7. Let $m \in \mathbb{Z}$ be an integer. If X satisfies

$$X' = z^{-q} a(z) X, \quad q \geq 2, a(z) \in \mathbb{C}[[z]], a(0) \neq 0,$$

then $z^m X$ satisfies the same equation where $a(z)$ is replaced by $a(z) + m z^{q-1}$. In particular, the order $q \geq 2$ of the pole $z = 0$ does not change.

PROPOSITION 3. Consider a differential system of the form (19) with $A_0(z) \neq 0$ and $v_z(A_0) \neq 0$. If

$$\exists c \in \mathbb{Z} \text{ such that } \forall k \in \mathbb{N}^*, v_z(A_k) \geq ck + v_z(A_0), \quad (20)$$

then the normalization $X \rightarrow z^c X$ yields a new differential system of the form (19) where p is replaced by $p - v_z(A_0)$, $v_z(A_0) = 0$, i.e., $A_0(0) \neq 0$, and $v_z(A_k) \geq 0$, for all $k \geq 1$.

PROOF. This can be deduced easily from the computation in the proof of Lemma 3. $\square$

Note that if $v_z(A_0) = 0$ but there exists $k_0 \in \mathbb{N}^*$ with $v_z(A_{k_0}) < 0$, then the proof of Lemma 3 also shows that, if we have (20), we can perform a normalization $X \rightarrow z^c X$, $c \in \mathbb{Z}^*$ to get a new system with coefficients having all non-negative valuation.

Finally, since (20) is always satisfied when we consider systems of the form (19) where $A(z, X)$ has entries in the ring $\mathcal{B}$ defined by (6) (see the proof of Lemma 3), we have:

THEOREM 4. The algorithm developed in Section 4 can be applied to compute a formal fundamental matrix of solutions of all systems of the form (19) with $A(z, X) \in \mathbb{M}_n(\mathcal{B})$.

REMARK 6. In the case where $A_0(z) = 0$ in System (19), then we can still use a normalization to come down to a system satisfying $p < q$. Indeed, let $\nu > 0$ denote the X -adic valuation of $A(z, X)$, then the normalization $X \rightarrow z^m X$ with $m = \lfloor (p - q)/\nu \rfloor + 1$ yields a new system with $p < q$. Note that, when $A_0(z) = 0$, one can also apply the change of variables $Z = XY$ to obtain a new system for Z satisfying $A_0(z) = z^{-q} a(z) I_n \neq 0$.

Let us illustrate the purpose of this section on an example.

EXAMPLE 2. Let X satisfy an equation of the form (1b), i.e., $X' = z^{-q} a(z) X$ and consider the scalar equation

$$Y' = -z^{-(q+1)} \left(\sum_{k=0}^{+\infty} A_k X^k \right) Y, \quad (21)$$

with

$$A_0 = z^q, \quad \forall k \geq 1, \quad A_k = z^{-(k-1)} (z^{q-1} - a(z)).$$

With the notation of the paper, we then have $p = q + 1 > q$. Clearly $\inf_{k \in \mathbb{N}} v_z(A_k) = -\infty$ so that our equation does not have coefficients in the ring $\mathcal{A}$ defined by (5). However, we can check that $v_z(A_k) \geq -k + 1$, for all $k \in \mathbb{N}$ so that the coefficients are in the ring $\mathcal{B}$ defined by (6) (since (6) is satisfied with $\alpha = -1$ and $\beta = 1$). This further implies that the condition (20) is fulfilled. Now $v_z(A_0) = q > 0$ so that Proposition 3 ensures that there exists $c \in \mathbb{Z}^*$ such that the normalization $X \rightarrow z^c X$ yields a new equation of the form $Y' = -z^{-1} \sum_{k=0}^{+\infty} \tilde{A}_k X^k$ with $\tilde{A}_0 = 1$ so that $\tilde{A}_0(0) \neq 0$, and $\tilde{A}_k \in \mathbb{M}_n(\mathbb{C}[[z]])$. From the proof of Lemma 3, since $v_z(A_0) = q > \beta = 1$, we can choose $c = \lfloor \alpha + \beta - v_z(A_0) \rfloor = -q$. The resulting equation $\mathcal{S}_{\tilde{A}, a}^{1, q}$ can now be handled by our algorithm in the case $p < q$ studied in Subsection 4.1. Theorem 1 then implies that we can compute a formal power series $T = 1 + T_1 X + T_2 X^2 + \dots$, with $T_k \in \mathbb{C}[[z]]$ that reduces the equation to the leading equation $y' = -z^{-1} \tilde{A}_0 y = -z^{-1} y$ that admits the trivial solution $y = z^{-1}$. Here we find that the coefficients T_k are given by $T_k = z^{(q-1)k}$. A formal fundamental solution of (21) can thus be written as

$$Y = \left(\sum_{k=0}^{+\infty} z^{(q-1)k} (z^{-q} X)^k \right) z^{-1} = z^{-1} \sum_{k=0}^{+\infty} z^{-k} X^k.$$

The latest power series is an expansion of $(z - X)^{-1}$ which is a solution of (21).

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